

Infections- Viral Edition (HIV, HSV, HPV, & TORCH Viruses)

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DNA viruses and retroviruses

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Session Objectives

- Explain how viruses function as obligate intracellular parasites, utilizing and manipulating host cellular machinery to support their replication and persistence.
- Compare the key biological features of retroviruses and DNA viruses, including their mechanisms of persistence, latency, and genome integration within host cells.
- Describe the clinical presentation, transmission routes, tissue tropism, and disease outcomes of viral infections, and evaluate the strategies retroviruses and DNA viruses use to evade and subvert host immune responses.
- Analyze how viral genome structure and replication strategies influence viral pathogenesis, latency, oncogenesis, diagnosis, prevention, and therapeutic interventions.

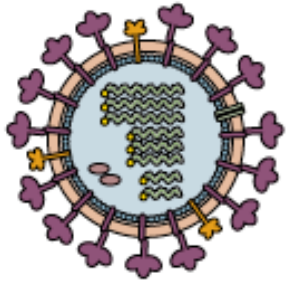
TORCH_eS

- **T – Toxoplasmosis** (*Toxoplasma gondii*)
- **O – Other infections** (includes Syphilis, Varicella-Zoster, Parvovirus B19, Hepatitis B, HIV, and Zika)
- **R – Rubella** (German measles)
- **C – Cytomegalovirus (CMV)**
- **H – Herpes Simplex Virus (HSV)**
- **e – Epstein-Barr virus / Enteroviruses** (sometimes omitted or grouped under "Others")
- **S – Syphilis** (or sometimes Group B Streptococcus, though syphilis is also frequently placed in the "Others" category)

Bonus: HPV – (not a torch)

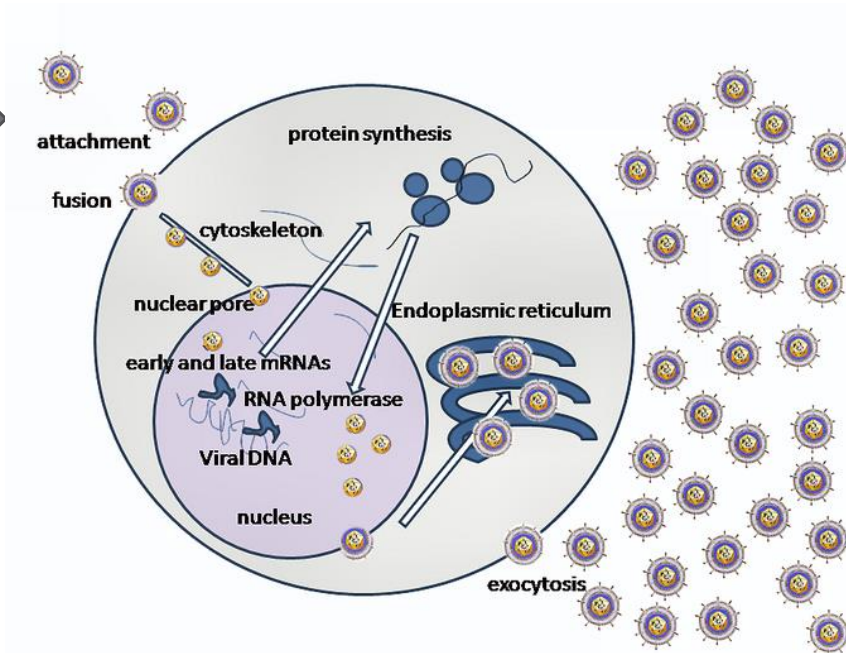


What is a virus?



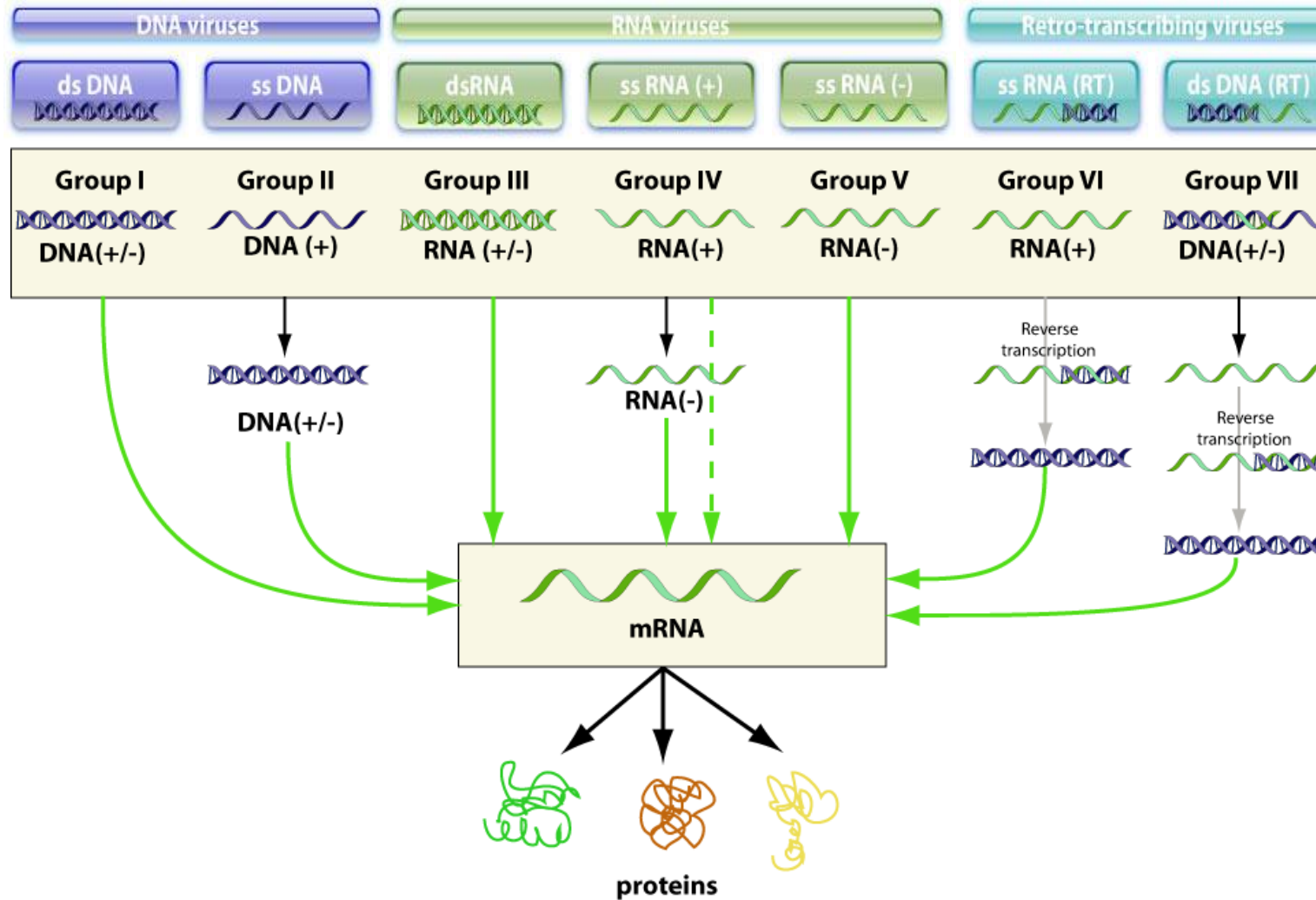
Delivery System

- Protein coat
- Nucleic acid core

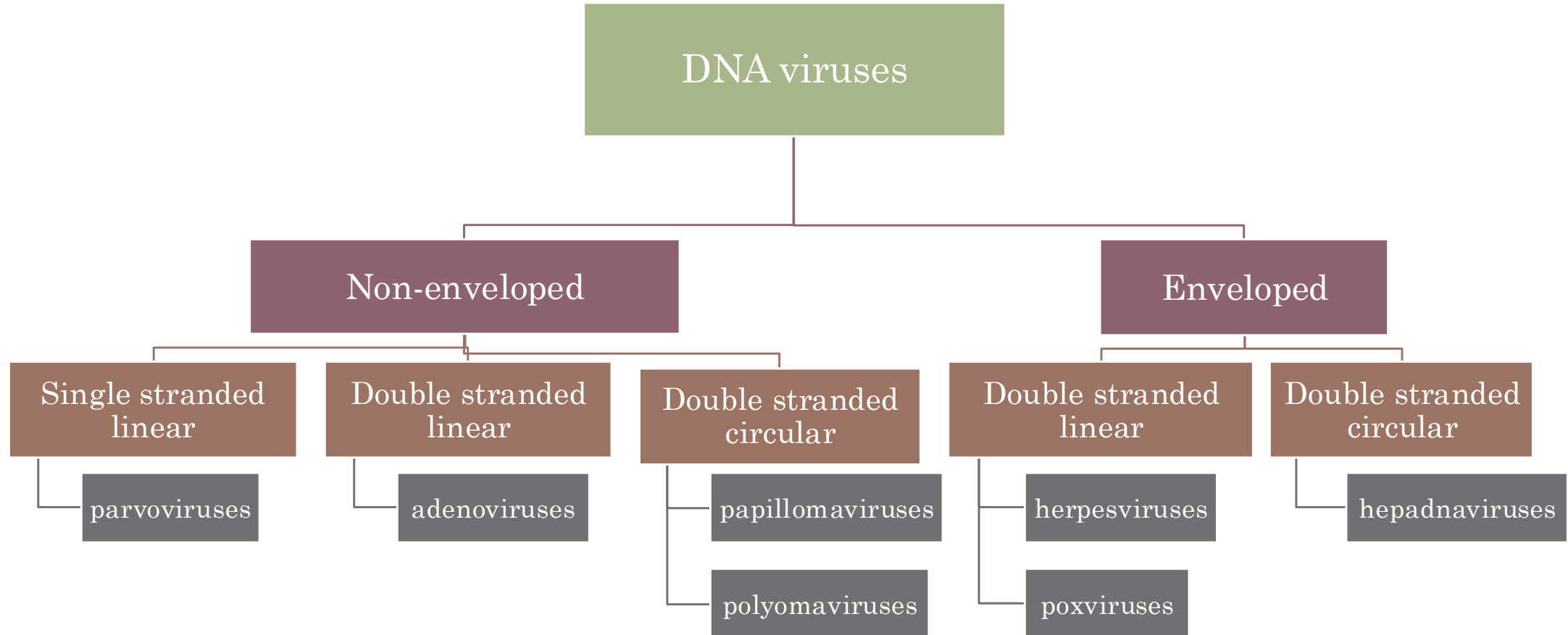


- Payload
 - A set of instructions for replication

Baltimore – Genome organization

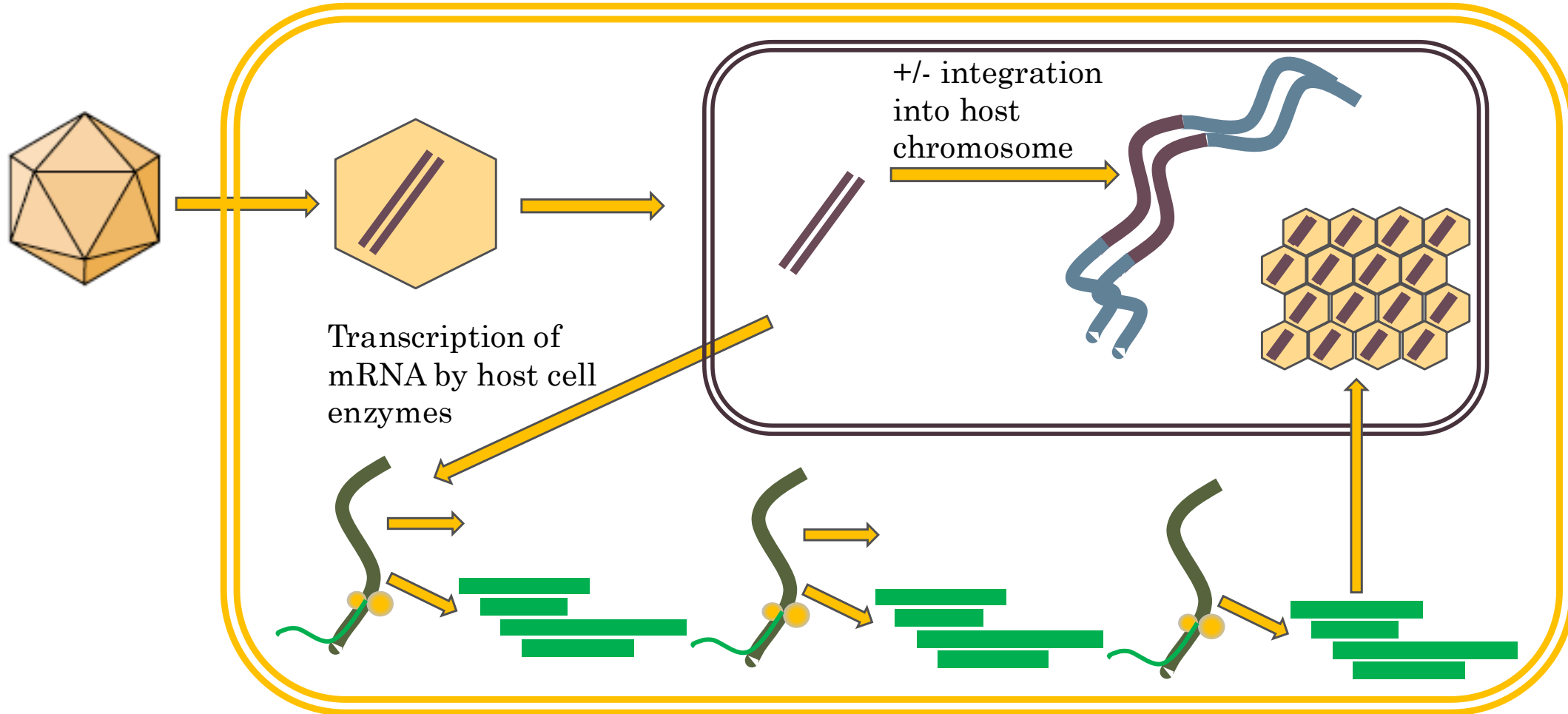


Virus classification- DNA



DNA viruses

Exception:
parvovirus
Single stranded



Complex transcription and translation cycle – IE, E, L

DNA viruses: Key points

- DNA viruses are HHAPPPP
- All double stranded except parvovirus (ssDNA)
- All replicate in the nucleus except poxvirus

Hepadna, Herpes, Adeno, Papaloma, Parvo, Pox, Polyoma



- You can get herpes in the backseat of a CHEVrolet (CMV, HSV, EBV, VZV)
- HepaDNAviruses = Hepatitis B virus



DNA enveloped viruses

- Herpesviruses

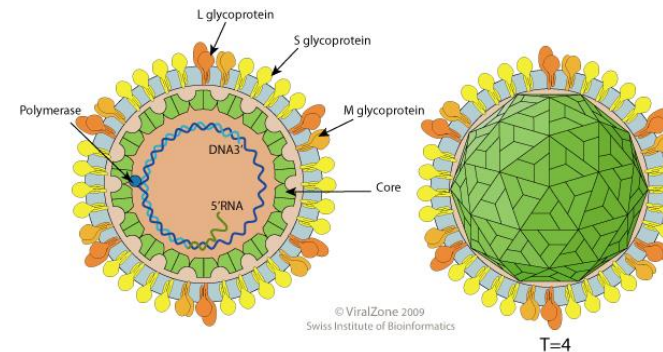
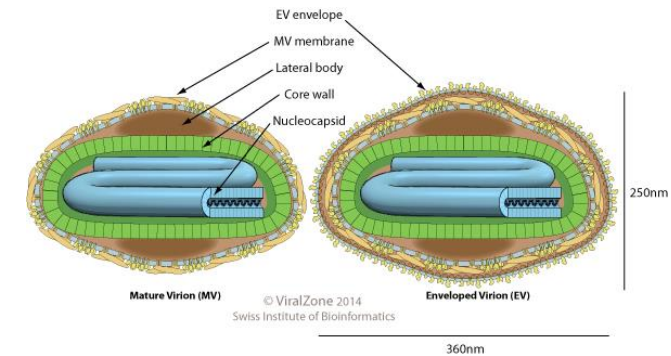
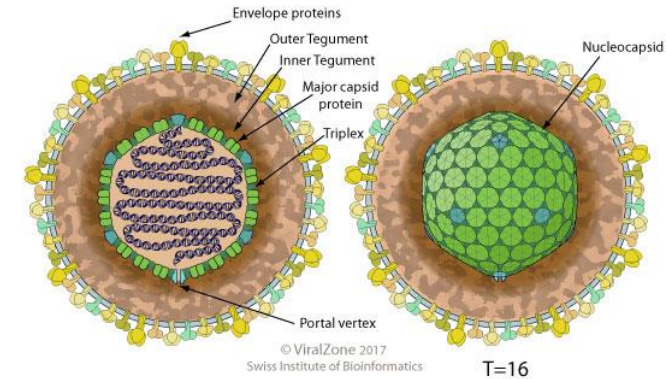
- Human alphaherpesvirus 1 and 2 (Herpes simplex viruses 1 and 2)
- Human alphaherpesvirus 3 (Varicella-Zoster virus)
- Human betaherpesvirus 5 (Cytomegalovirus)
- Human gammaherpesvirus 4 (Epstein-Barr virus)
- Human gammaherpesvirus 8 (Kaposi's Sarcoma-Associated herpesvirus)

- Poxviruses

- Smallpox virus
- Molluscum contagiosum virus

- Hepadnaviruses

- Hepatitis B virus



Herpesviruses key points

- Herpesviruses (>130 known viruses)
 - Human alphaherpesvirus 1 and 2 (Herpes simplex viruses 1 and 2)
 - Human alphaherpesvirus 3 (Varicella-Zoster virus)
 - Human betaherpesvirus 5 (Cytomegalovirus)
 - Human gammaherpesvirus 4 (Epstein-Barr virus)
 - Human gammaherpesvirus 8 (Kaposi's Sarcoma-Associated herpesvirus)
- All have an icosahedral core surrounded by an envelope
- Genome is double stranded and huge
- Replicate in the nucleus
 - They are the only viruses that obtain their envelope from the nuclear membrane

TAXONOMY

Group I: dsDNA viruses

Order: ***Herpesvirales***

Family: ***Herpesviridae***

Subfamily: *Alphaherpesvirinae*

Genus: *Iltovirus*

Mardivirus

Scutavirus

Simplexvirus

Varicellovirus

Subfamily: *Betaherpesvirinae*

Genus: *Cytomegalovirus*

Muromegalovirus

Roseolovirus

Proboscivirus

Subfamily: *Gammaherpesvirinae*

Genus: *Lymphocryptovirus*

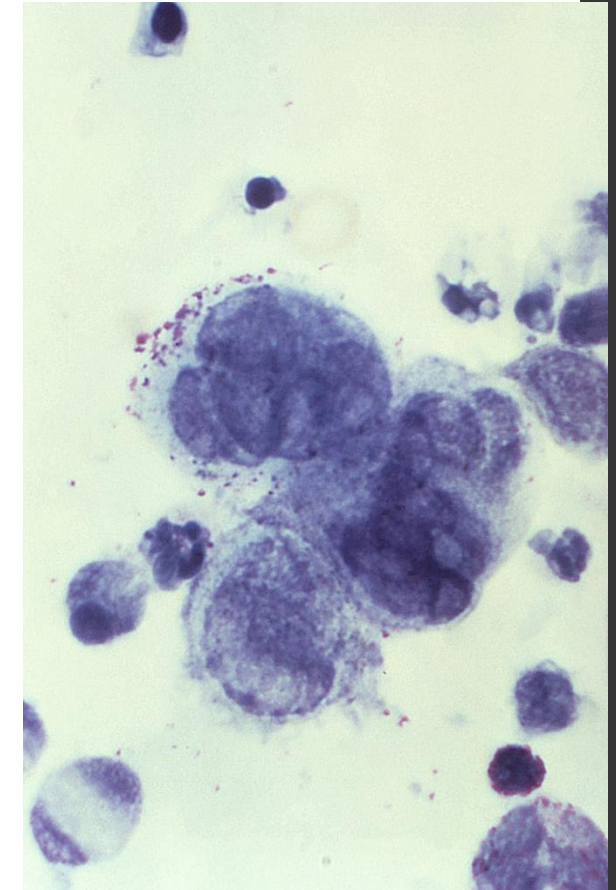
Rhadinovirus

Macavirus

Percavirus

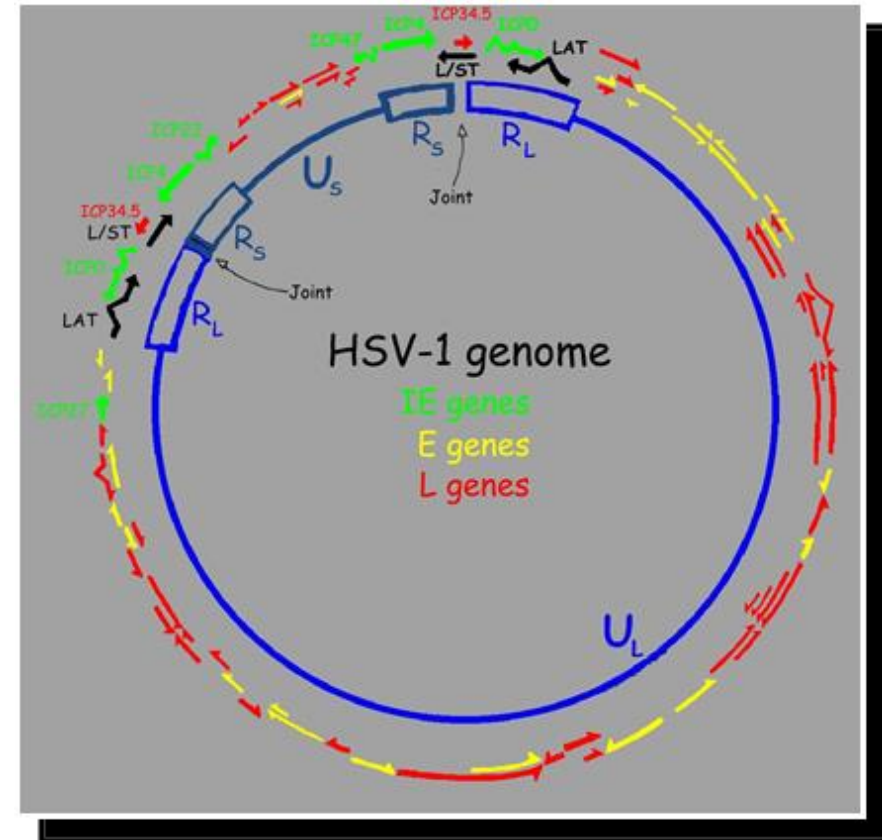
Herpesviruses key points

- Cause latent infections
 - herpesviruses have evolved different strategies to inhibit the host adaptive immune response ... MHC stability, TAP protein blocking, prevent IRF3 activation
 - Transcription is limited to a few **latency-associated-transcripts (LATs)**, which suppress viral replication
- Vesicular rash: HSV1, HSV2, and VZV
- Multinucleated giant cells: HSV1, HSV2, VZV, and CMV
 - Tzanck smear (pictured)
- Tropism
 - Alphaherpesviruses: HSV1, HSV2, and VZV infect epithelial cells and cause latent infections in neurons
 - Betaherpesviruses: CMV and HHV6 infect and become latent in a variety of cell types
 - Gammaherpesviruses: EBV, and KSAV (HHV8) infect and become latent in lymphoid cells
- Cancer – EBV and KSHV



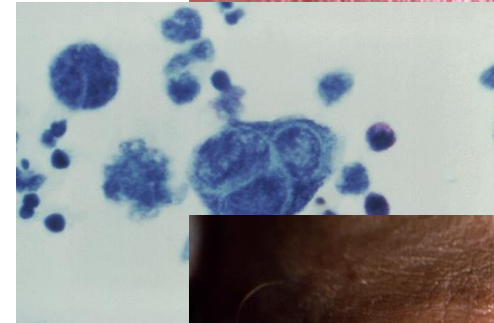
Herpes simplex viruses (HSV): replication

- Cell receptors: heparin sulfate, nectin, others
- Fusion of envelope and cell membrane
- vDNA enters the nucleus and is circularized
- **5 IE proteins:**
 - VP16 interacts with host cellular transcription factors
 - Host RNA polymerase transcribes IE genes
 - IE proteins regulate the synthesis of E proteins (e.g., vDNA polymerase that replicates the viral genome)
- **25 E proteins:**
 - 5 nucleotide synthesis enzymes (thymidine kinase)
 - 7 DNA replication proteins (HSV DNA polymerase)
- **45 L proteins:**
 - 5 capsid proteins (VP5)
 - 15 tegument proteins (VP16)
 - 11 glycoproteins (gD)
 - 6 involved in nucleocapsid assembly (protease, endonuclease)



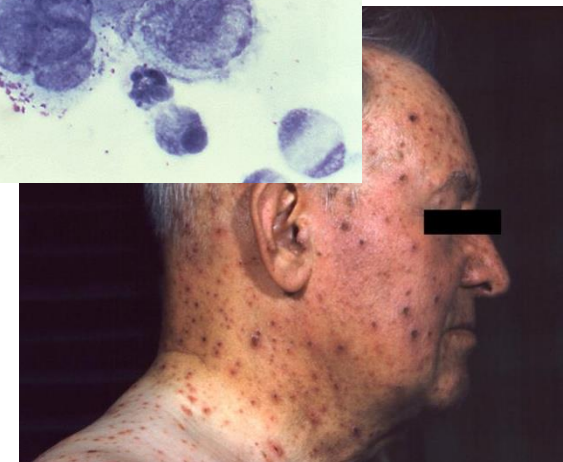
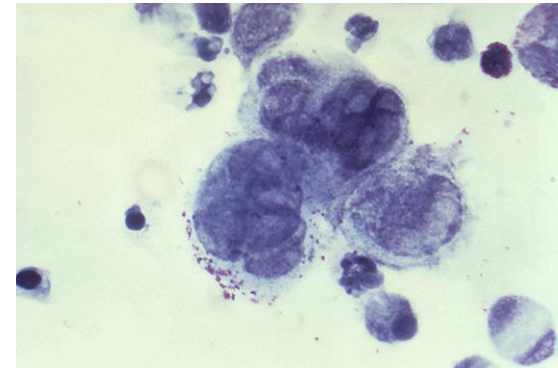
Herpes simplex viruses (HSV): diseases

- Pathogenesis
 - Virus replicates in the skin and mucous membranes
 - Migrates and becomes latent in the sensory ganglion (HSV1 – trigeminal; HSV2 – lumbar and sacral)
- Transmission
 - HSV1 – saliva, face
 - HSV2 – sexual contact, genitals
 - Viral shedding during symptoms and when asymptomatic
- Diseases
 - Vesicular lesions – serous fluid
 - HSV1 – gingivostomatitis, herpes labialis (cold sores) keratoconjunctivitis, and encephalitis (usually in adults)
 - HSV2 – genital herpes, neonatal encephalitis
- Immunity – Cell Mediated Immunity and Ab?
- Dx: serology, PCR, Tzanck smear
- Treatment – Acyclovir ([ViPR](#))



Varicella-Zoster virus (VZV): disease

- Varicella (chickenpox); Zoster (shingles)
- Transmission
 - By respiratory droplets and direct contact
- Pathogenesis
 - Highly contagious
 - Infects the mucosa of the URT and then spreads by blood to the skin
 - Latency in dorsal root ganglia
 - Reactivation – reduced cellular immunity, zoster
- Diseases
 - Varicella
 - Incubation 14-21 days
 - Papulovesicular rash – trunk > head > extremities
 - Zoster
 - **Painful** vesicles along the course of sensory nerves, head and trunk
 - Immunocompromised patients can be life threatening
- Treatment – rarely needed, acyclovir
 - If resistant Foscarnet, famciclovir and valacyclovir
- Vaccine – Live attenuated
 - Varivax (varicella) and Zostavax (zoster)



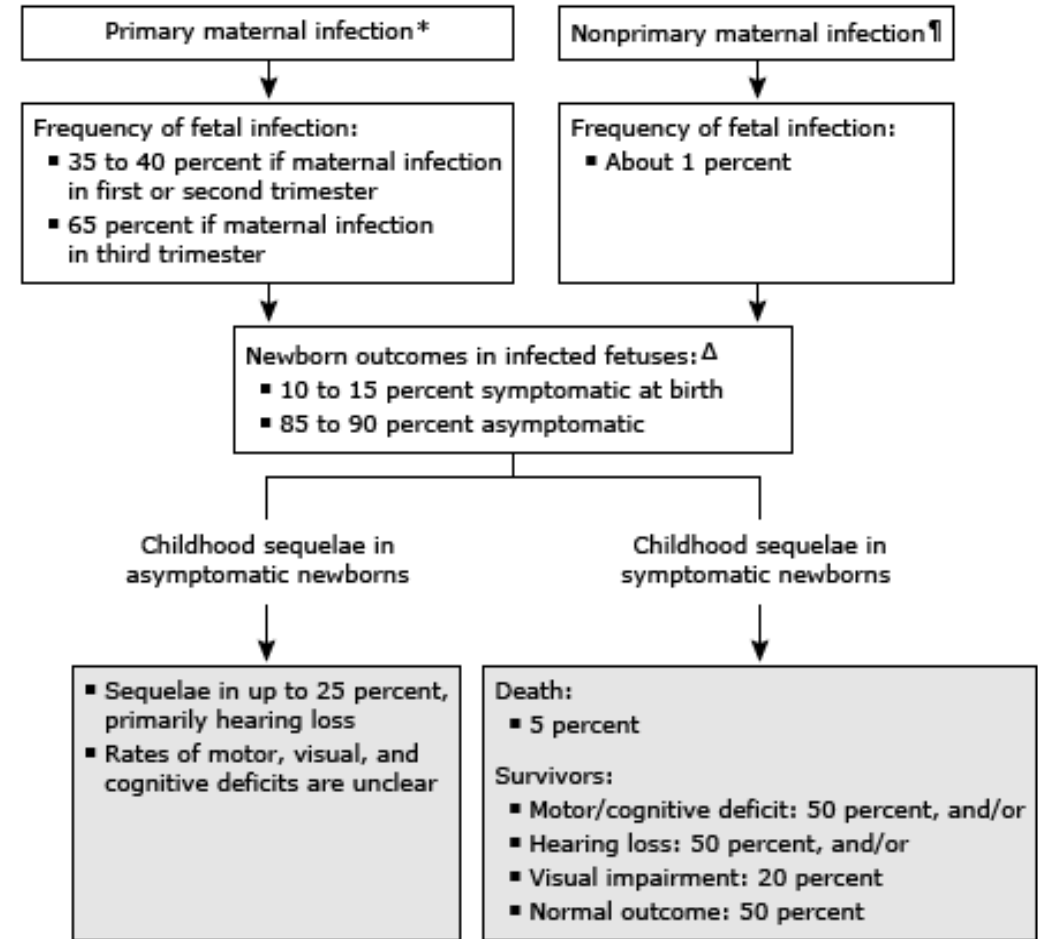
Cytomegalovirus (CMV): disease

- Transmission

- CMV is transmitted by a variety of modes – vertical, birth canal, breast milk, saliva, sexually, blood
- CMV enters a latent state in monocytes
- Reactivated when CMI is low

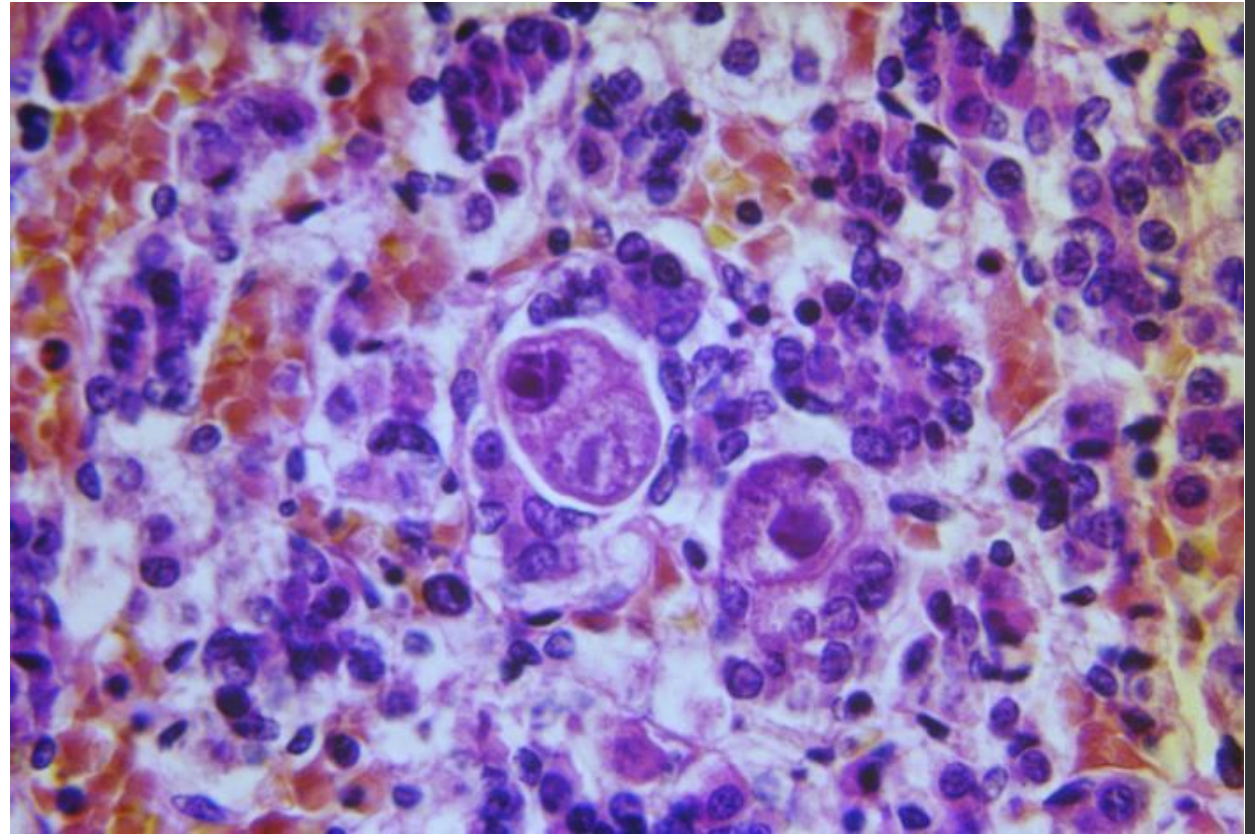
- Disease

- Infection of children and adults is usually asymptomatic
- Immunocompetent adults - **heterophil-negative mononucleosis**
 - Fever, lethargy, and abnormal lymphocytes on blood smear
- **Greatest concern is primary infection of the mother during the first trimester of pregnancy**
 - Clinical findings include microcephaly, ventriculomegaly, chorioretinitis, jaundice, hepatosplenomegaly, thrombocytopenia, and petechiae
 - The mortality rate among symptomatic newborns is approximately 5 percent
 - 50 to 60 percent of survivors develop serious long-term neurologic morbidity (eg, progressive hearing and/or visual impairment, motor/cognitive impairment)



Cytomegalovirus (CMV): diagnosis

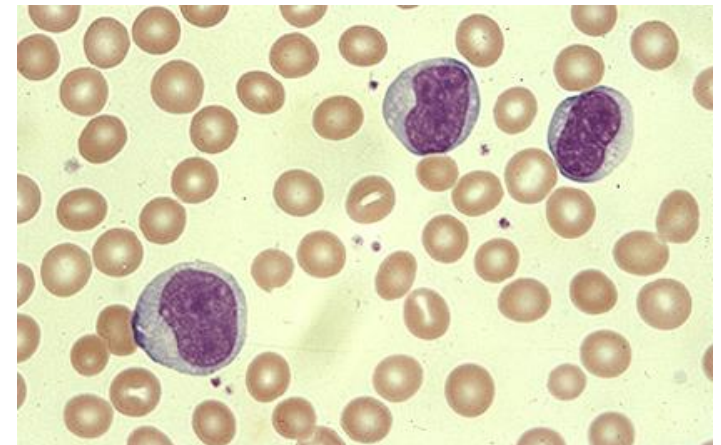
- Photomicrograph of a section of pituitary gland tissue revealed the presence of cytomegalic inclusion cells, also referred to as owl-eye inclusion cells, in a case of cytomegalic inclusion disease, caused by the cytomegalovirus
- Serology – 4X increase
- IFA – pp65 antigen in blood leukocytes
- PCR



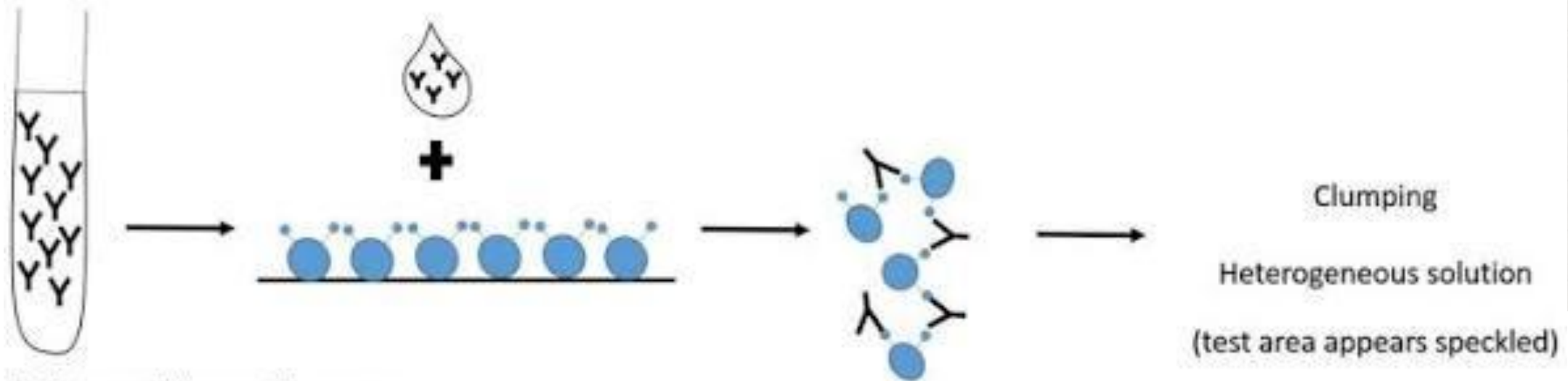
Source CDC

Epstein-Barr virus (EBV): disease

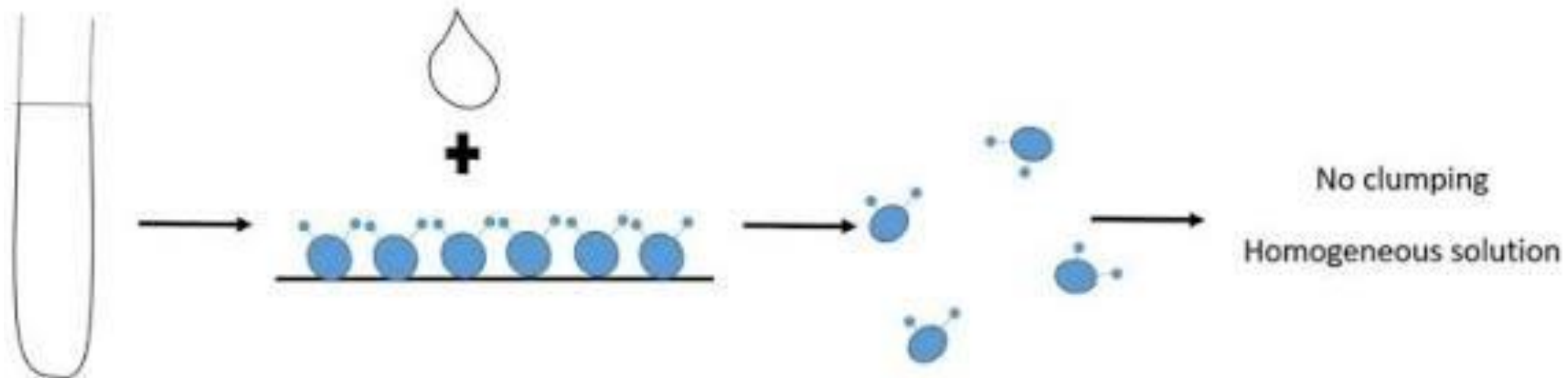
- EBV causes infectious mononucleosis
- Transmission: saliva “kissing disease”
 - Primary active infection
 - Reactivation of latent infection
- Pathogenesis
 - Infects cells in the oropharynx
 - Viremic spread to B lymphocytes
- Disease
 - Infections within the first few years of life is asymptomatic
 - Clinical appearance is fever, sore throat, lymphadenopathy, splenomegaly
 - Spontaneous recovery in ~ 3 weeks
 - X-linked lymphoproliferative syndrome T and NK cell deficiencies. 75% mortality
 - Associated with Burkitt’s lymphoma, B cell lymphoma, T cell lymphoma, Hodgkin lymphoma, and nasopharyngeal carcinoma
- Diagnosis
 - Hairy leukoplakia, occurs in immune deficient patients (HIV)
 - Lymphocytosis – 30% abnormal lymphocytes, CTL
 - Heterophil antibody positive (Monospot)
 - Serology – VCA IgM or IgG



A Heterophile positive serum



B Heterophile negative serum

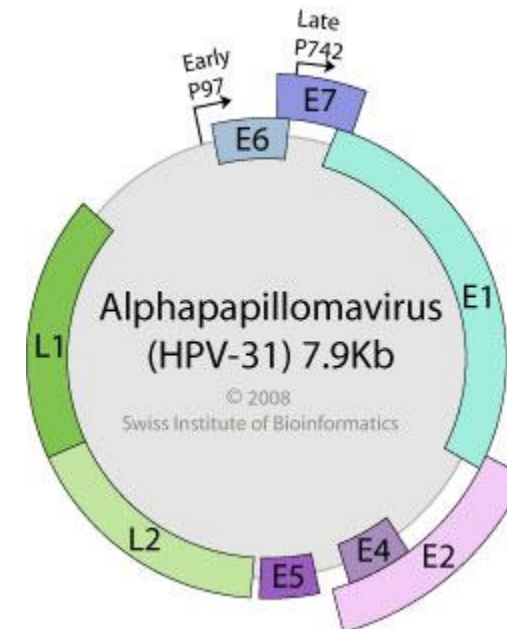
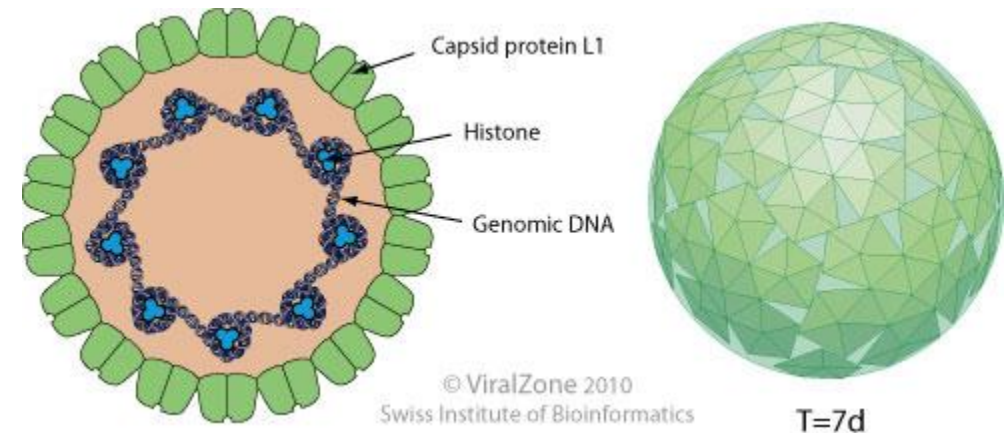


Hepadnavirus (HBV):

- Not a torch

Papillomaviruses: virion

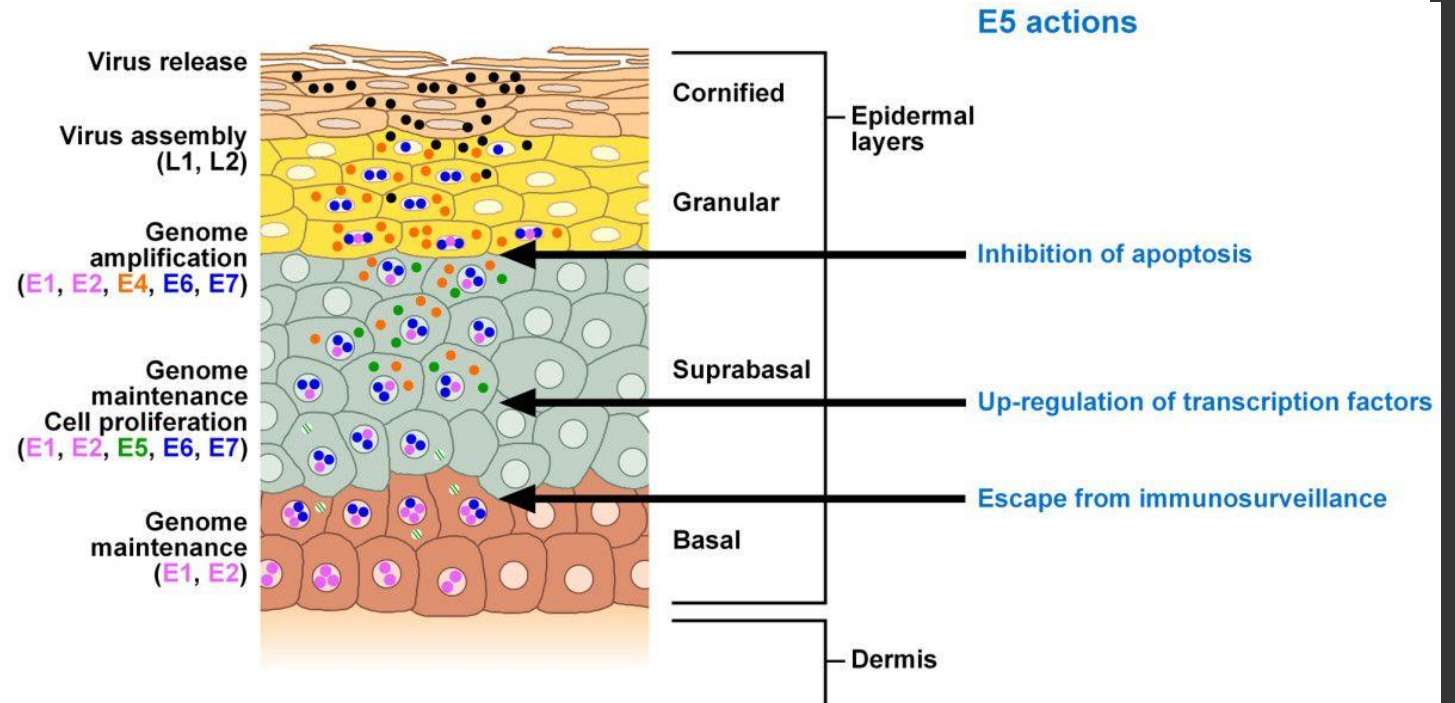
- Non-enveloped, icosahedral capsid
- Circular dsDNA
- Genome encodes
 - Early genes (E1-E7)
 - Synthesis of mRNA and replication of progeny DNA
 - Late genes (L1 and L2)
 - Structural proteins
- E6 and E7 inactivate p53 and Rb, respectively
Cell Cycle proteins!

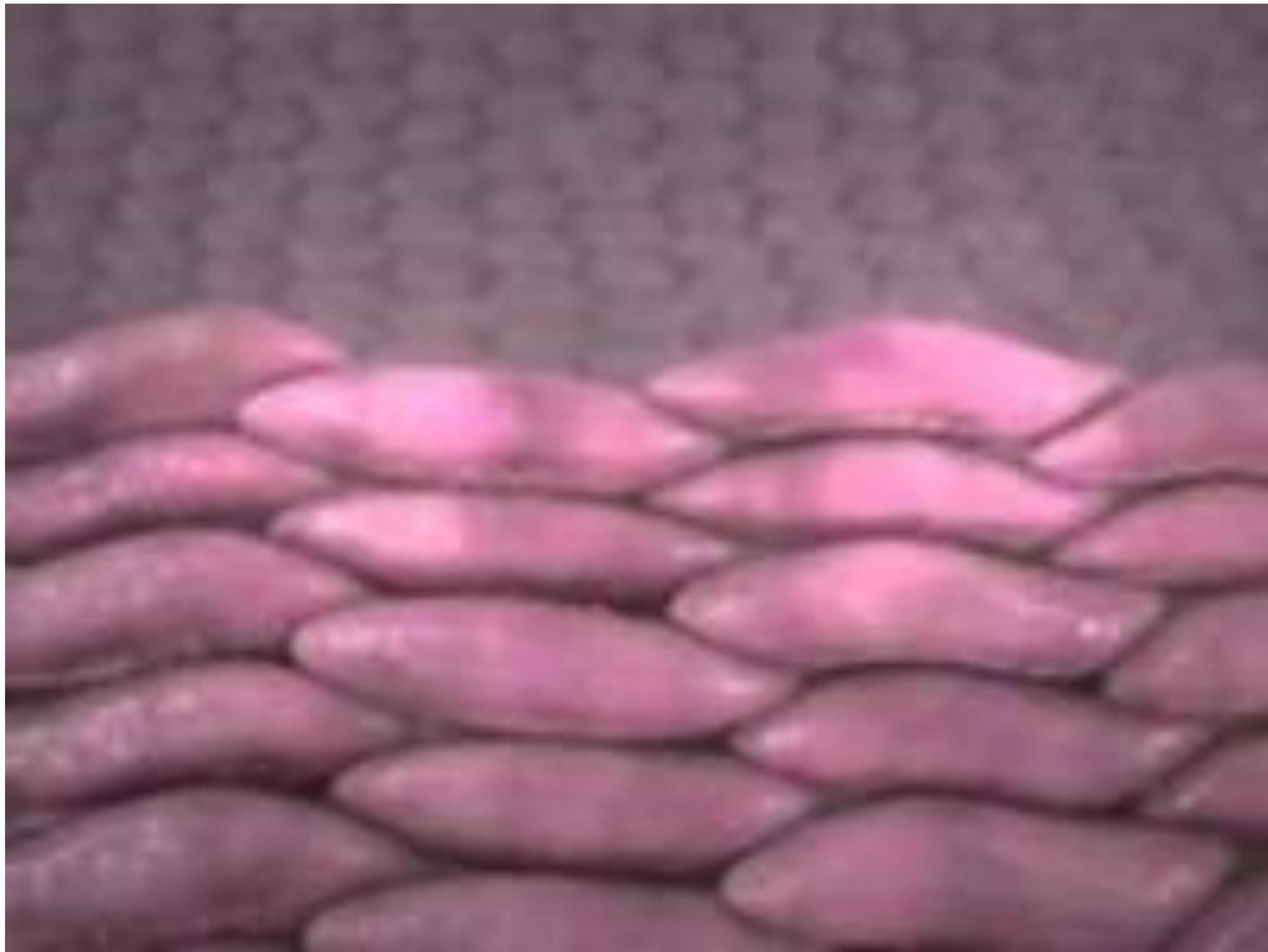


Papillomaviruses: replication

- [HPV Replication](#)

- The virus gains access to the basal layer of the tissue through micro-abrasion
- The initial progeny genomes are maintained as episomes in the nucleus
- Only early genes are expressed in the basal cells
- As the cells differentiate, substantially higher levels are mature virions are detected
- Virus is released from terminally differentiated cells



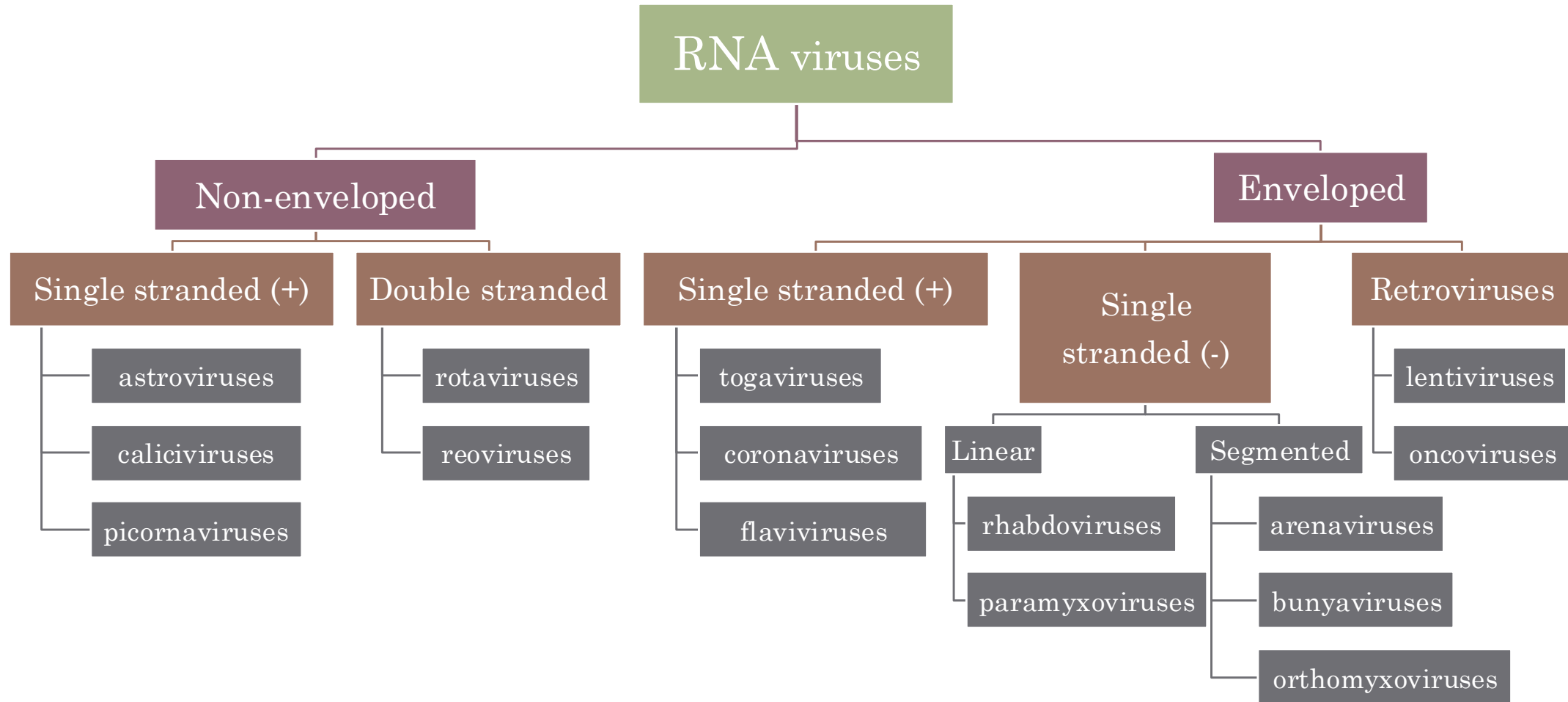


Papillomaviruses: disease

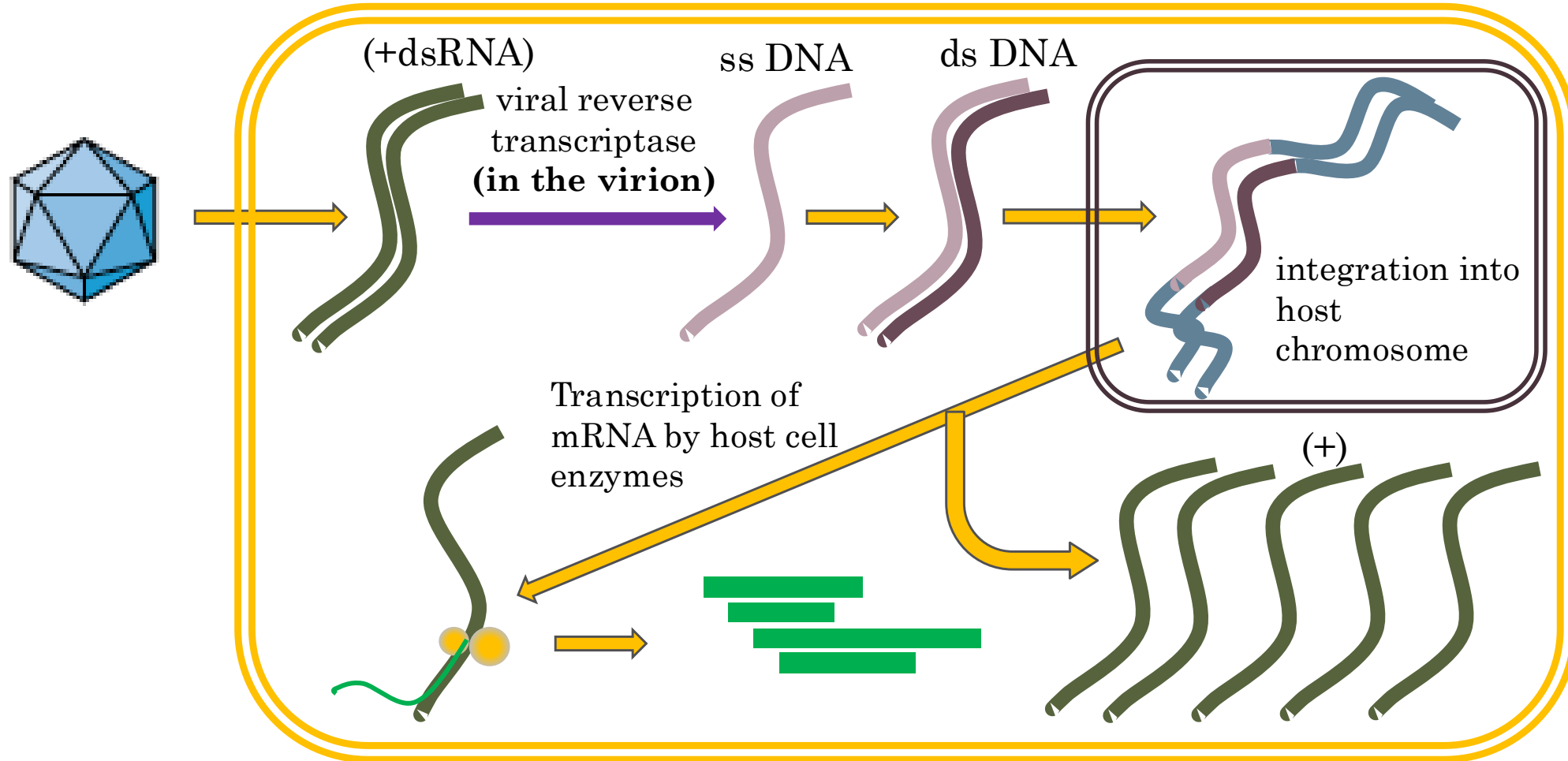
- >100 species, in 39 genera
 - HPV-1 thru HPV-4 infect the skin
 - HPV-6 and HPV-11 cause genital warts
 - HPV-16 and HPV-18, among others, infect the genital tract
- Transmission
 - Skin-to-skin contact
 - Sexual
- Clinical findings
 - Skin and plantar warts
 - Genital warts
 - Carcinoma of the cervix, penis, anus
 - E6 (p53) and E7 (Rb)
- Vaccines
 - Very effective in preventing carcinoma of the cervix, anal carcinoma, and genital warts
 - Gardasil 9 – recombinant vaccine against 9 types of HPV, contains HPV-6, 11, 16, 18, 31, 33, 45, 52, and 58



Virus classification- RNA



Retroviruses Replication



Retroviruses

- Medically important retroviruses
 - Deltaretroviruses
 - HTLV-1
 - HTLV-2
 - Lentiviruses
 - HIV-1
 - HIV-2

Retroviridae

● Unclassified retroviridae

Orthoretrovirinae

● Alpharetrovirus

● Betaretrovirus

● Gammaretrovirus

● Deltaretrovirus

PTLV-1

PTLV-2

PTLV-3

BLV

● Epsilonretrovirus

● Lentivirus

HIV-1

HIV-2

Spumaretrovirinae

● Simiispumavirus Renamed

● Bovispumavirus New

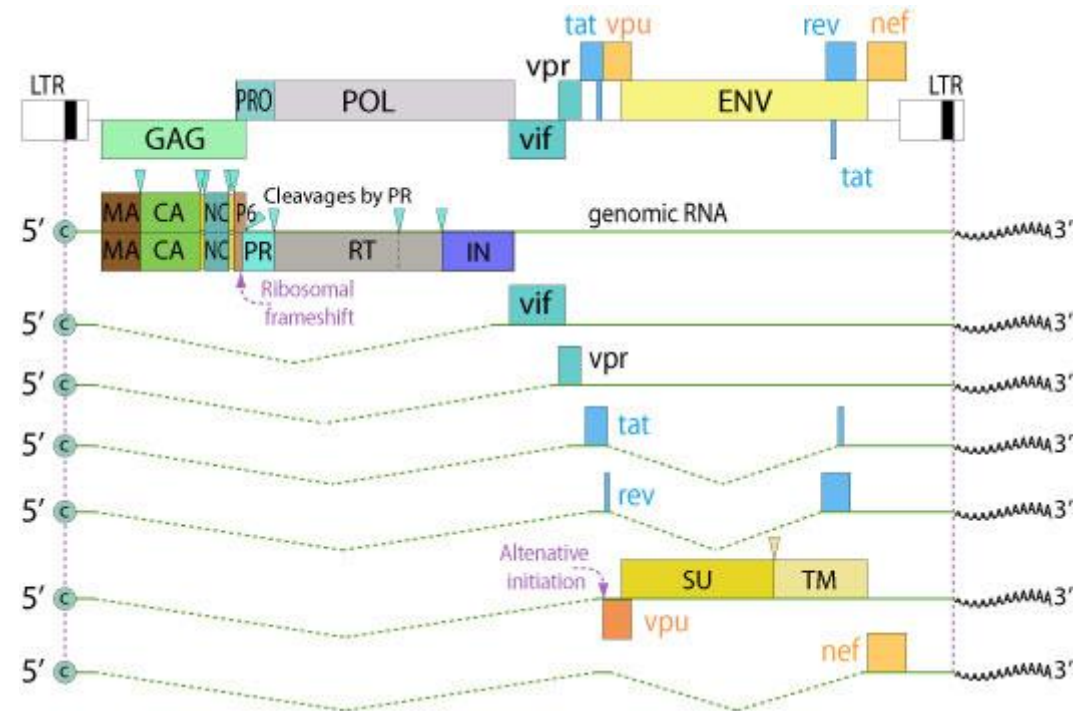
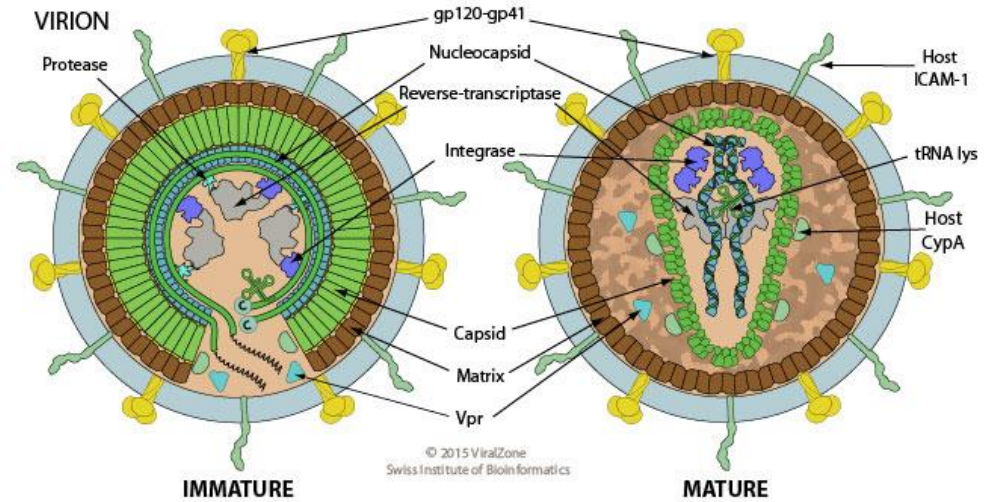
● Equispumavirus New

● Felispumavirus New

● Prosimiispumavirus New

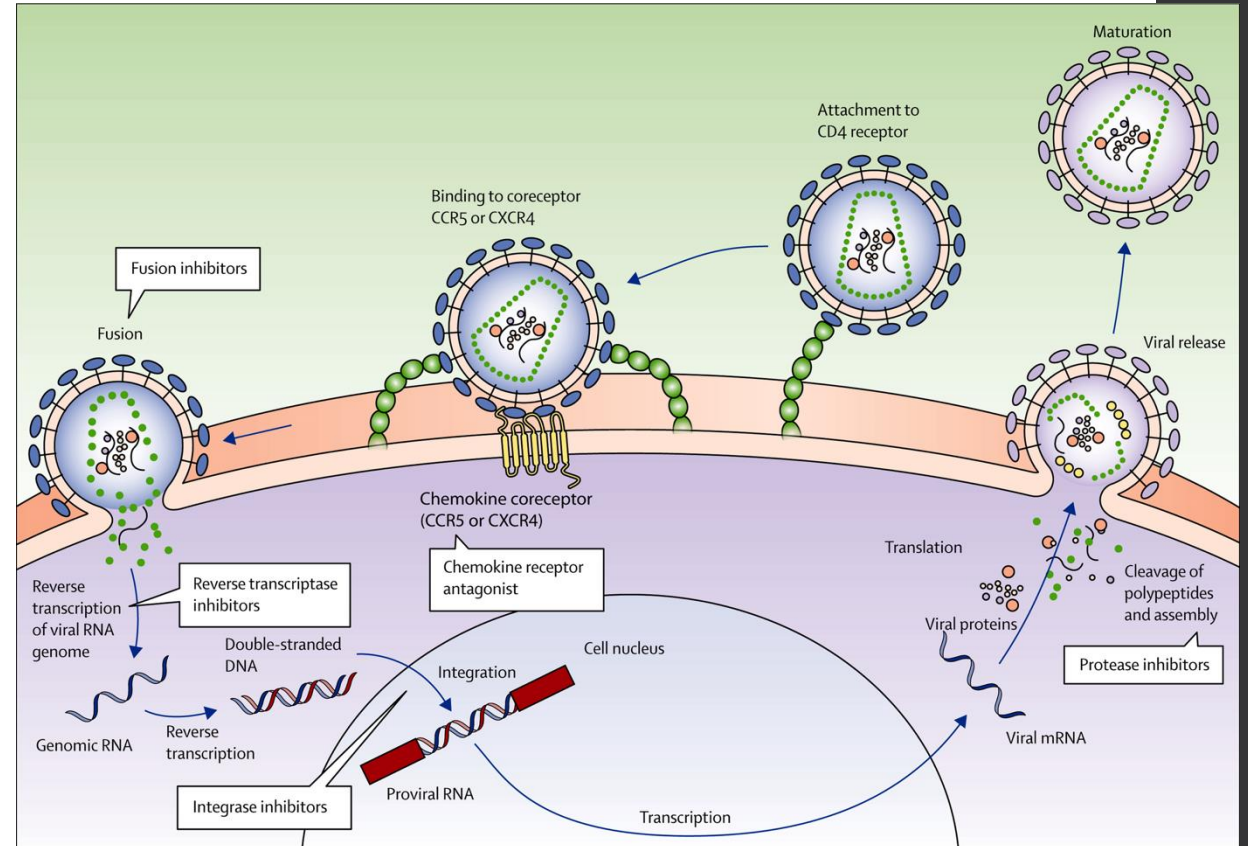
Lentivirus

- linear, diploid, ssRNA(+) genome
- Typical retrovirus genome arrangement with genes for structural proteins
 - GAG – nucleocapsid protein, p24
 - POL – reverse transcriptase, integrase, protease
 - ENV - gp160 > gp120 and gp41
- Regulatory genes
 - tat – repress synthesis of MHC-I
 - rev – controls passage of mRNA into the cytoplasm
 - nef – repress synthesis of MHC-I
 - vif – inhibits APOBEC3G (innate immune)
 - vpr
 - vpu

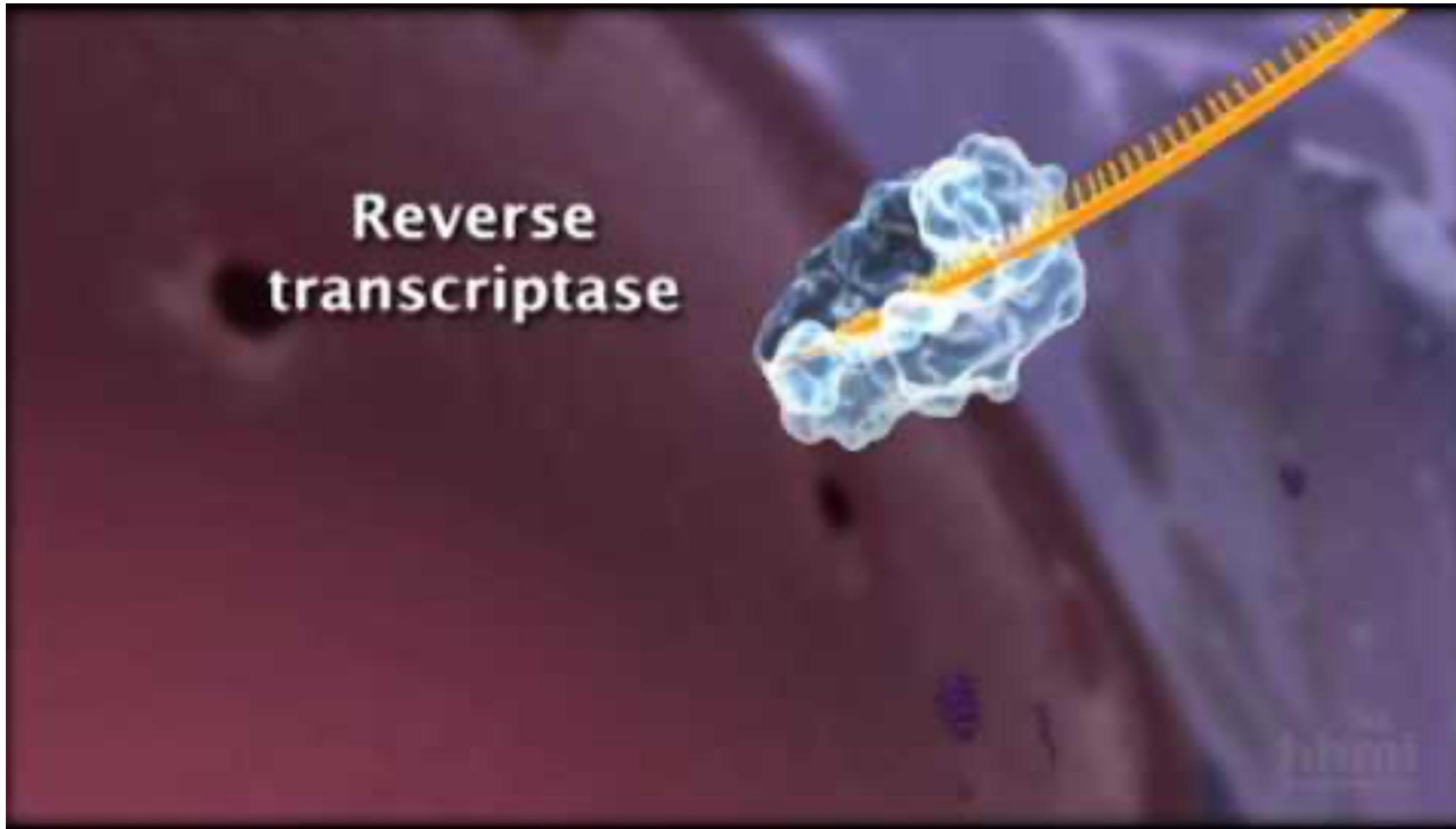


Lentivirus: replication

- The main target of HIV is activated CD4 T lymphocytes
- gp120 interacts with the CD4 protein and chemokine co-receptors CCR5 or CXCR4
- gp41 mediates fusion
- RT transcribes the genome RNA into double stranded DNA
- vDNA integrated into the host cell genome (integrase – viral encoded!)
- Viral mRNA is transcribed from the pro-viral DNA
- gag and pol polyproteins are cleaved by the viral protease
 - gag > (p24 and p17)
 - pol > reverse transcriptase, integrase, protease
- gp160 protein is cleaved by cellular protease



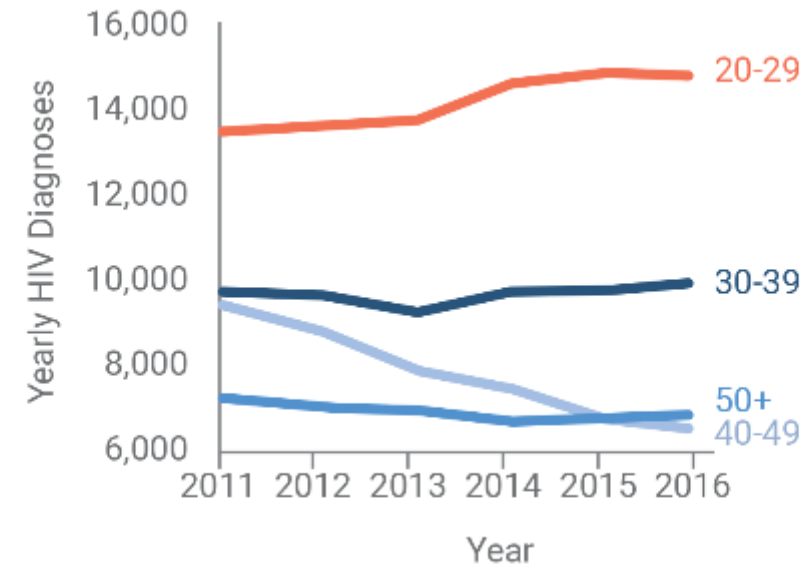
HIV Replication



HIV: transmission

- 1.1 million people in the United States (2015), 34M worldwide
 - The annual number of new diagnoses in US declined by 5% from 2011 to 2015
 - Africa, Asia and Latin America have the highest rates of new infections
 - Approximately 630,000 people have died of AIDS in the US since 1981
- Transmission
 - Sexual
 - Blood
 - Perinatal Birth vs Breast Milk
- U = U

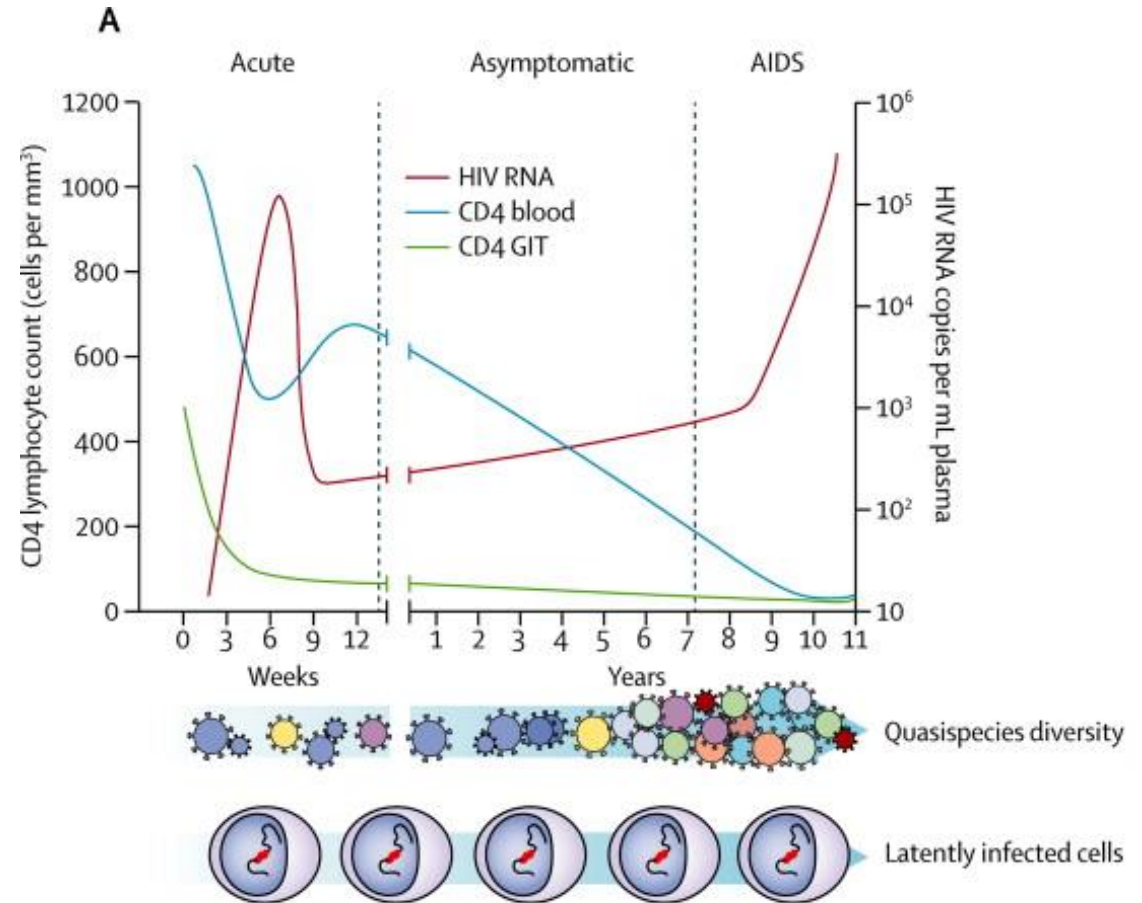
Age
HIV diagnosis



- HIV infections **increased by 10%** among those **20-29 years old** between 2011 and 2016¹⁵

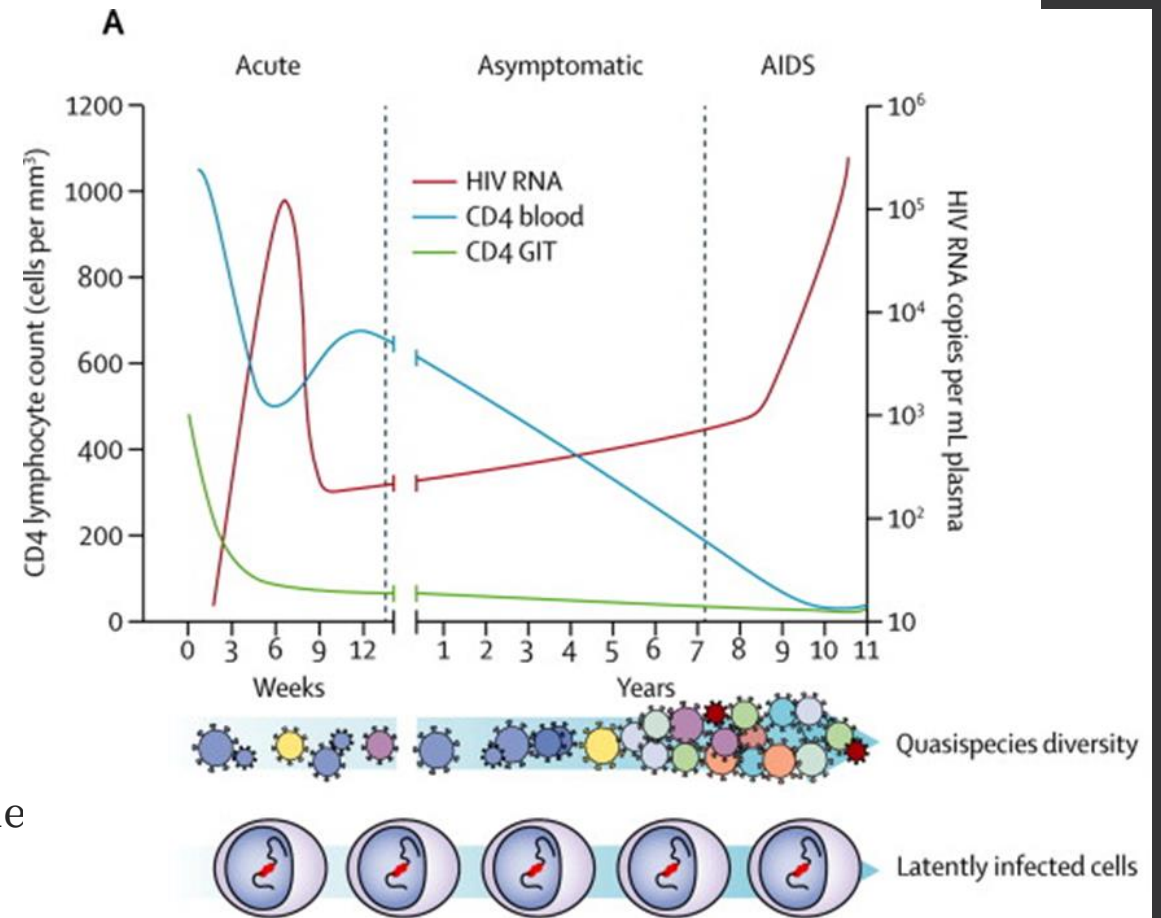
HIV: Pathogenesis & Immunity

- The first hallmark of infection is a rapid increase in HIV replication
- Induction of inflammatory response
 - NK cells
 - CD8+ CTL
 - Neutralizing antibody
- Progressive depletion of CD4 T cells
 - Massive depletion of activated CD4 T cells in the GI tract
 - CD8+ T cells fail to proliferate
- AIDS



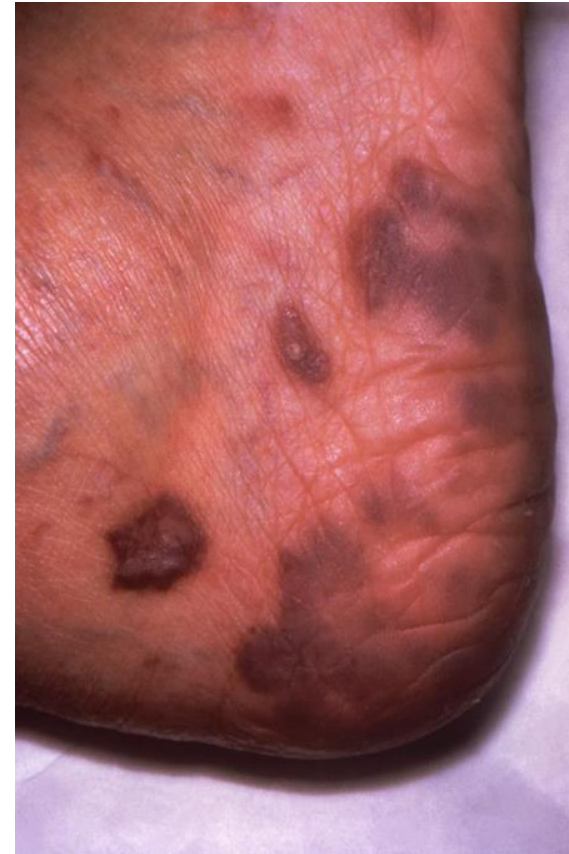
HIV: Clinical Findings (2)

- Three stages of HIV infection
 - Acute
 - Begins 2-4 weeks after infection
 - Fever, lethargy, generalized lymphadenopathy, maculopapular rash
 - Leukopenia, normal CD4
 - High viral load, decreases later in the acute stage
 - Resolves in 2-3 weeks
 - Viral load “set point”
 - Latent
 - Long latent phase, typically years
 - Asymptomatic, lower viral load
 - Virus replication and sequestration in LN
 - Aids related complex (ARC): persistent fever, fatigue weight loss, and lymphadenopathy
 - Immunodeficiency



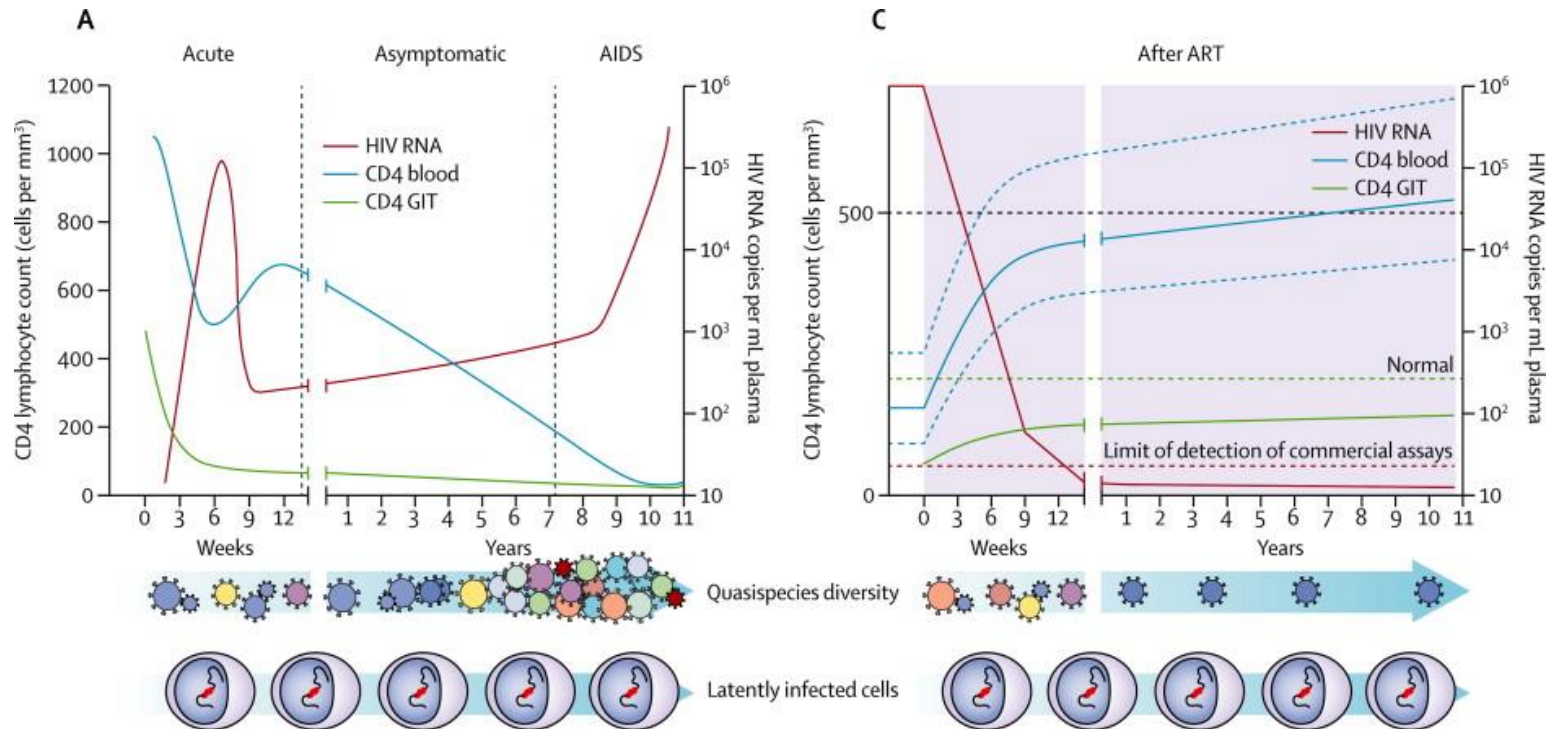
HIV: Clinical Findings

- Three stages of HIV infection
 - Immunodeficiency
 - CD4+ T cells < 200/ μ L
 - AIDS
 - Opportunistic infections – *Pneumocystis jiroveci*
 - Kaposi's sarcoma
- Diagnosis
 - ~~p24 Ab or Ag ELISA~~
 - ~~Western blot (p24, gp41) is often used for confirmation~~
 - RT-PCR – viral RNA in plasma, inform treatment decisions
 - CD4+ T cell counts - inform treatment decisions
 - **5th Generation Elisa**
 - **HIV -1, HIV-2 (antibodies) p24 (antigen)**



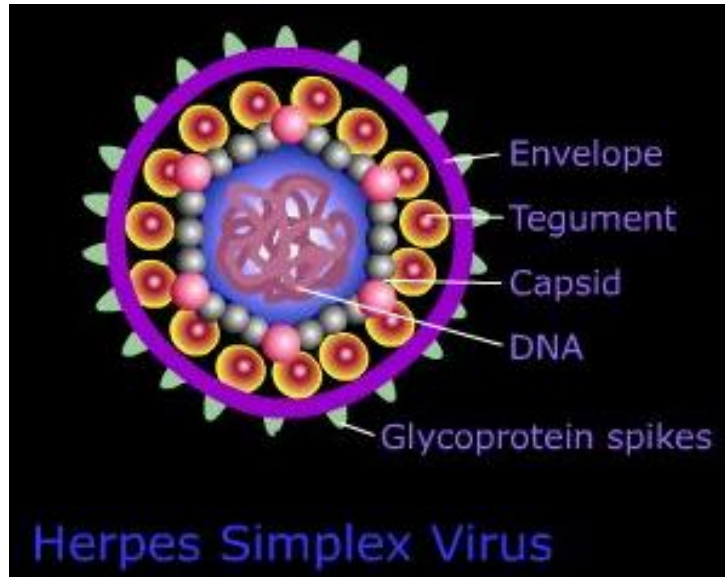
HIV: treatment

- Treatment, three specific goals
 - Restore immunologic function by increasing CD4+ T cells
 - Decrease opportunistic infections
 - Decrease viral load



- Antiretroviral drugs
 - Entry inhibitors
 - Reverse transcriptase inhibitors
 - Protease inhibitors
 - Integrase inhibitors

Genital Herpes Simplex (HSV)



HSV1 - Oral Herpes



HSV2 - Genital Herpes



HSV Infection – General Info

- Virus
- HSV-1 – usually oral infection (herpes labialis or cold sores)
- HSV-2 – usually genital infection (herpes genitalis)
- Transmission
 - HSV-1 via oral secretions or sores on the skin, can be spread through kissing or sharing objects such as toothbrushes or eating utensils.
 - HSV-2 generally via sexual contact with someone who has a genital HSV-2 infection.
 - It is important to know that both HSV-1 & HSV-2 can be spread even if sores are not present.
 - It can also be transmitted to newborn during a vaginal delivery
- Risks
 - Being a women. Women are more likely to have genital herpes than are men. The virus is sexually transmitted more easily from men to women than it is from women to men.
 - Having multiple sexual partners. Each additional sexual partner raises your risk of being exposed to the virus that causes genital herpes.

HSV Infection

- HSV can go through periods of being dormant
- Attacks (or outbreaks) can be brought on by:
 - General illness (colds, etc)
 - Fatigue or lack of sleep
 - Physical or emotional stress
 - Immunosuppression due to AIDS or such medications as chemotherapy or steroids
 - Trauma to the affected area, including sexual activity
 - Menstruation
- Symptoms
 - A Blister or Multiple Blisters on or around affected areas -- usually the mouth, genitals, or rectum.
 - The blisters break, leaving tender sores.

HSV Screening & Prevention

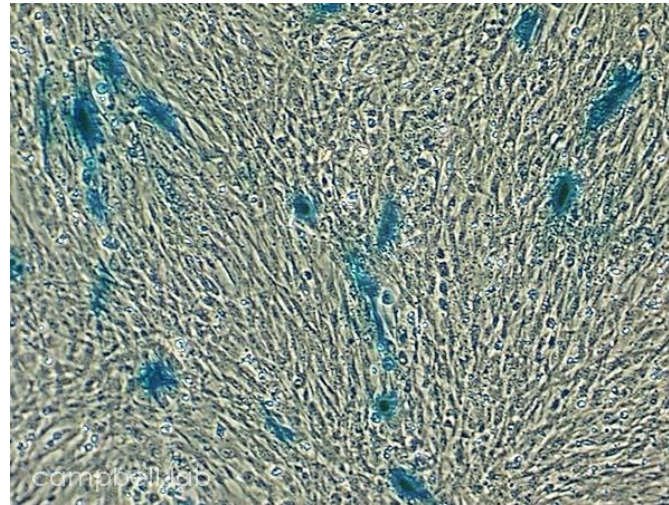
- Screening - CDC does not currently recommend routine testing in asymptomatic individuals or in pregnancy
- Use condoms – may help but not if lesions are not covered with the condom
- Limit sexual partners or Practice abstinence
- Refrain from intercourse if partner has active herpetic outbreak until it is cleared
- Male medical circumcision demonstrated reduced acquisition in 2 African countries
 - This benefit seemed to transfer to women as well
- PrEP with daily oral Tenofovir Disoproxil Fumarate and Emtricitabine decreases risk for HSV-2 acquisition by 30% in heterosexual partnerships
- Pericoital intravaginal Tenofovir 1% gel also decreases the risk for HSV-2 acquisition among heterosexual women

HSV Counseling - CDC

- Testing for genital herpes can be useful when someone has genital symptoms that might be related to herpes
- Testing helps by confirming the diagnosis and allowing the provider to counsel about what to expect in the future, the role of treatment, and what preventive measures should be taken to reduce transmission to current and future partners
- Testing can also be useful when someone has a sex partner with history of genital herpes to see if they are infected
- If uninfected, then counseling about ways to reduce risk of acquiring genital herpes infected should be offered
- Herpes blood testing may be useful for people seeking a full STI evaluation, especially for people with multiple sex partners
- Testing is recommended for people with symptoms, because treatment is available to partially control the symptoms of herpes
- In addition, persons with symptomatic genital herpes who take daily antiviral medication can reduce the risk of spreading HSV-2 to their sex partners by 50%
- Asymptomatic viral shedding is most frequent during the first 12 months after acquiring HSV-

HSV – Diagnosis

- Physical exam showing typical herpetic ulcerations
- Viral culture
 - “Gold” standard for HSV laboratory testing is cell culture
 - ELVIS™ (Enzyme Linked Viral Induced System) is another cell culture test but results are ready in ONE day



- Polymerase chain reaction (PCR) test which can determine whether one has HSV-1 or 2 or both
 - Chlamydia transport medium
- Blood test like IgM and IgG

HSV – Treatment

DISEASE	RECOMMENDED REGIMEN	ALTERNATIVE REGIMEN
Genital Herpes Simplex		
First clinical episode of genital herpes ⁷	acyclovir 400 mg orally 3x/day for 7–10 days ⁸ OR famciclovir 250 mg orally 3x/day for 7–10 days OR valacyclovir 1 gm orally 2x/day for 7–10 days	
Suppressive therapy for recurrent genital herpes (HSV-2)	acyclovir 400 mg orally 2x/day OR valacyclovir 500 mg orally 1x/day ⁹ OR valacyclovir 1 gm orally 1x/day OR famciclovir 250 mg orally 2x/day	
Episodic therapy for recurrent genital herpes (HSV-2) ¹⁰	acyclovir 800 mg orally 2x/day for 5 days OR acyclovir 800 mg orally 3x/day for 2 days OR famciclovir 1 gm orally 2x/day for 1 day OR famciclovir 500 mg once, FOLLOWED BY 250 mg 2x/day for 2 days OR famciclovir 125 mg 2x/day for 5 days OR valacyclovir 500 mg orally 2x/day for 3 days OR valacyclovir 1 gm orally 1x/day for 5 days	
Daily suppressive therapy for persons with HIV infection	acyclovir 400–800 mg orally 2x–3x/day OR famciclovir 500 mg orally 2x/day OR valacyclovir 500 mg orally 2x/day	
Episodic therapy for persons with HIV infection	acyclovir 400 mg orally 3x/day for 5–10 days OR famciclovir 500 mg orally 2x/day for 5–10 days OR valacyclovir 1 gm orally 2x/day for 5–10 days	
Daily suppressive therapy of recurrent genital herpes in pregnant women ¹¹	acyclovir 400 mg orally 3x/day OR valacyclovir 500 mg orally 2x/day	

The Thymidine Kinase (TK)

- **Mechanism of Selectivity:**

- **Acyclovir** remains the "prototypic" antiherpetic. Its selectivity is governed by a viral "gatekeeper" enzyme:

1. **Initial Activation:** Acyclovir must be monophosphorylated by **viral-encoded TK**. This enzyme is **only present in infected cells**, **preferential** activation occurs in **HSV-infected** cells.

2. **Host Completion:** Host cellular kinases then convert the monophosphate into the active acyclovir triphosphate, which competitively inhibits viral **DNA polymerase** and **terminates** viral DNA **elongation**.

Valacyclovir

- **Pharmacokinetic Optimization**

- Acyclovir has poor oral bioavailability; valacyclovir is its L-valyl ester prodrug with **greater oral bioavailability**, achieving acyclovir levels comparable to IV acyclovir and allowing less frequent oral dosing.

- **Dosing Strategies**

Distinguish between three clinical scenarios:

- **First Episode:** Intensive, high-dose treatment for initial primary infection.
- **Episodic Recurrent:** "On-demand" treatment initiated by the patient at the first sign of a prodrome (tingling/burning).
- **Chronic Suppressive:** Daily therapy indicated for patients with severe/frequent ≥ 6 **episodes per year**. This strategy significantly reduces both the frequency of outbreaks and the risk of transmission to seronegative partners.

Safety Profile: Crystalline Nephropathy

- IV Acyclovir is excreted unchanged in the urine. If the drug concentration exceeds its solubility in the renal tubules, it precipitates.
- **Nephroprotection**
 - High-dose IV Acyclovir can cause **Crystalline Nephropathy** (obstructive uropathy). To mitigate this risk, ensure the patient is aggressively **hydrated** with IV normal saline to maintain high tubular flow rates.

HSV – Pregnancy Treatment

- Pregnancy
 - Both animal & human data on Acyclovir exposure during pregnancy, including the 1st trimester, suggest that this drug is safe at all stages of pregnancy
 - ACOG & CDC recommend treatment from 36 weeks of gestation until delivery with
 - Acyclovir 400 mg orally three times daily or
 - Valacyclovir 500 mg orally twice a day

HSV – Complications

- Other STIs, including HIV
- Newborn infection exposed to the virus during the birthing process which may result in brain damage, blindness or death for the newborn
 - Pregnant women with HSV can deliver vaginally UNLESS they have genital sores at the time of delivery at which point a C-section will be performed
- Bladder problems due to sores associated with genital herpes causing inflammation of the urethra and the resultant swelling can close the urethra for days, requiring the insertion of a catheter
- Meningitis (rare)
- Proctitis, particularly in men who have sex with men

TORCH Pathogens

Cytomegalovirus and Maternal-Fetal Considerations

- Cytomegalovirus (CMV) is a high-stakes TORCH pathogen. Pharmacological management focuses on **inhibiting viral DNA polymerase**, though the pathways to activation differ.
- **Ganciclovir / Valganciclovir:** Requires initial monophosphorylation by the viral **UL97 kinase**. Major dose-limiting toxicity is **neutropenia** (bone marrow suppression).

Alternative /resistant CMV agents

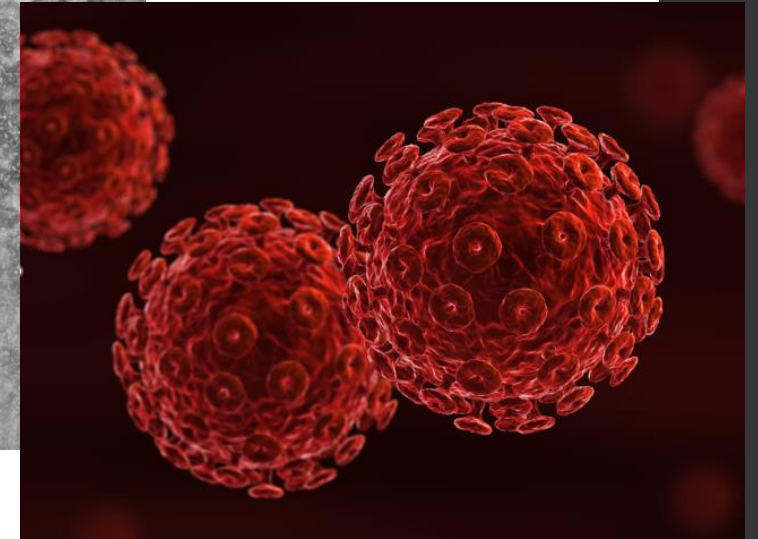
- **Foscarnet: A pyrophosphate derivative.** Unlike -cyclovirs, it **does not require phosphorylation** by any kinase. It directly binds the pyrophosphate-binding site of viral DNA polymerase.
 - **Adverse Effect:** Nephrotoxicity. It "chelates" divalent cations, leading to severe **hypocalcemia** and **hypomagnesemia**.
- **Cidofovir: A cytosine nucleotide analog.** Because it is already a nucleotide (possesses one phosphate), its activation is **independent of viral enzymes**.
 - **Nephroprotection:** Cidofovir is highly nephrotoxic. It **must** be coadministered with **Probenecid** (to block tubular secretion) and **Normal Saline** hydration.

Pregnancy Safety

Managing viral infections in pregnancy requires balancing maternal health against fetal teratogenicity.

Treatment	Pregnancy Status	Clinical Justification
TCA / BCA	Preferred	Minimal systemic absorption; localized caustic effect.
Acyclovir	Generally Safe	Extensive safety data for HSV management.
Podophyllotoxin	Contraindicated	Mitotic inhibitor; risk of fetal harm.
Ribavirin	CONTRAINDICATED	Contraindicated in pregnancy; highly teratogenic.

HIV



HIV Screening & Exposure Testing

Recommendation Summary USPSTF

- HIV screening recommended for all sexually women, women with IV drug use, and women in prenatal care (T1 & T3 if high risk) using opt-out approach (ACOG & CDC)
 - Blood tests
 - Antibody-Antigen combination tests – 18-45 days after exposure
 - NAT tests for RNA – 10-33 days after exposure
 - Buccal testing (antibody testing) – 23-90 days after exposure

Population	Recommendation	Grade
Pregnant persons	The USPSTF recommends that clinicians screen for HIV infection in all pregnant persons, including those who present in labor or at delivery whose HIV status is unknown.	A
Adolescents and adults aged 15 to 65 years	The USPSTF recommends that clinicians screen for HIV infection in adolescents and adults aged 15 to 65 years. Younger adolescents and older adults who are at increased risk of infection should also be screened. See the Clinical Considerations section for more information about assessment of risk, screening intervals, and rescreening in pregnancy.	A

HIV Transmission

- Intercourse – use condoms, PrEP
- Blood – needle exchange programs, follow best practices for needle usage in healthcare, PrEP
- Maternal-Fetal (Birth & Breast Milk) – testing, antiviral therapy

HIV PrEP

- [Preventing HIV with PrEP | HIV | CDC](#) (video 1 min)
 - No interaction between hormonal birth control and PrEP
- For **receptive anal sex (bottoming)**, PrEP pills reach maximum protection from HIV at about **7 days** of daily use.
- For **receptive vaginal sex** and **injection drug use**, PrEP pills reach maximum protection at about **21 days** of daily use.

Principles of Antiviral Selectivity and **HIV** Prevention

- Because viruses hijack host metabolic machinery for replication, the primary challenge is achieving a therapeutic index that spares the host cell.
- We have shifted from merely managing **opportunistic infections** to a strategy of **direct viral suppression** using **Antiretroviral Therapy (ART)**.
- This success relies on targeting **viral-specific enzymes**—such as **Reverse Transcriptase (RT)**, **Protease**, and **Integrase**—that have no direct functional counterparts in human physiology.

Nucleoside and Nucleotide Reverse Transcriptase Inhibitors (**NRTIs**)

- NRTIs, such as Emtricitabine (**FTC**) and Tenofovir, are the structural backbone of **Antiretroviral Therapy (ART)**.
- **Mechanism of Action:**
 - NRTIs are analogs of native ribosides that lack a **3'-hydroxyl group**.
 - To become active, they **must** undergo **intracellular phosphorylation** by host cellular enzymes into triphosphate (or diphosphate for nucleotides) forms.
 - Viral RT preferentially incorporates these active analogs into the growing viral DNA strand. Because the 3'-hydroxyl group is missing, the enzyme cannot form the requisite 3',5'-phosphodiester bond with the next incoming nucleoside. This results in the absolute **termination of DNA chain elongation**.

NRTIs

- **Selectivity and Toxicity:**

- While NRTIs have a higher affinity for HIV RT than for host DNA polymerases, **Mitochondrial DNA Polymerase/gamma** remains susceptible.
- This **cross-reactivity** is the primary driver -NRTIs have been associated with potentially fatal lactic acidosis and hepatomegaly with steatosis

Tenofovir Formulations (TDF vs. TAF)

- **Tenofovir** exists in two **prodrug** forms: Tenofovir disoproxil fumarate (**TDF**) and Tenofovir alafenamide (**TAF**).
 - Understanding their pharmacokinetic differences is essential.

Tenofovir Formulations

Feature	Tenofovir disoproxil fumarate (TDF)	Tenofovir alafenamide (TAF)
Circulating Plasma Levels	High (leading to systemic exposure)	Low (more stable in plasma)
Intracellular Levels	Lower	Higher (efficiently loaded into target cells)
Bone Density Impact	Significant decrease in BMD	Minimal impact
Renal Safety	Significant renal toxicity	Lower renal toxicity

Oral PrEP: TDF/FTC or TAF/FTC

Long-Acting Prophylaxis and Chelation

- Cabotegravir, dolutegravir, and bictegravir are integrase strand transfer inhibitors (INSTIs); long-acting injectable cabotegravir is used for HIV **PrEP**.
- Like all INSTIs and the endonuclease inhibitor **Baloxavir**, its mechanism involves the **binding** of viral enzymes **through** the **chelation** of **divalent** metal cations.
- Oral INSTIs (e.g., dolutegravir, bictegravir) and baloxavir interact with polyvalent cations (Ca, Mg, Fe, Al) in antacids/supplements; baloxavir should also be avoided with dairy products.
- Coadministration leads to drug chelation in the GI tract, significantly **reducing bioavailability**.

Additional HIV Agents

- Atazanavir (Protease inhibitor): inhibits HIV protease → prevents viral polyprotein cleavage and formation of mature infectious virions.
- Atazanavir: requires an acidic environment; take with food; may cause indirect hyperbilirubinemia/jaundice and PR prolongation.
- Cobicistat is a pharmacokinetic booster—not an antiviral—that inhibits CYP3A to increase exposure to selected antiretroviral drugs.
 - Major issue = drug interactions

Maternal HIV Management

- Preferred: bictegravir or dolutegravir + TAF/FTC, TDF/FTC, TAF/3TC, or TDF/3TC.
- Bictegravir and dolutegravir are preferred INSTIs in pregnancy; bictegravir is available as the single-tablet BIC/FTC/TAF regimen.
- Prenatal vitamins: iron/calcium can reduce oral INSTI absorption → separate administration by ≥ 2 hours.
- Avoid initiating cobicistat-containing regimens: pregnancy lowers drug exposure and has been associated with loss of virologic control.

Questions?

References

- [CDC](#)
 - <http://www.cdc.gov/STI/tg2015/2015-pocket-guide.pdf>
- [Webmed](#)
- [NIH](#)
- [AAFP](#)
- [Mayoclinic](#)